

Sketch-to-SQL with Execution-Guided Reinforcement Learning for Small Language Models

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1 Project Description

Text-to-SQL systems translate natural language questions into executable SQL queries over structured databases. Although large language models perform well with supervised fine-tuning (SFT), they often struggle with complex queries requiring reasoning over tables, joins, and aggregations. Recent reinforcement learning (RL) approaches suggest that optimizing models with execution-based rewards can improve reasoning ability without complex reward engineering.

In this project, we study whether introducing an intermediate sketch representation can improve RL-based post-training for small Text-to-SQL models. Instead of generating SQL directly from the input question and database schema, the model first produces a structured sketch describing key query components (tables, columns, joins, and aggregations), and then generates the final SQL conditioned on this sketch. We hypothesize that this intermediate structure helps the model organize its reasoning and improves the effectiveness of RL optimization.

Our training pipeline has two stages. First, we perform supervised fine-tuning on a dataset such as Spider, where sketches are automatically extracted from ground-truth SQL queries. The model learns to generate both sketches and SQL. Second, we apply reinforcement learning using Group Relative Policy Optimization (GRPO) with a reward based on SQL execution results.

We compare four systems: (1) Direct SQL with SFT, (2) Sketch-to-SQL with SFT, (3) Direct SQL with GRPO-based RL, and (4) Sketch-to-SQL with GRPO-based RL (our proposed method). Evaluation focuses on execution accuracy, SQL validity, and analysis of intermediate sketches to determine whether RL improves the model's reasoning decisions. We also examine potential reward-hacking behaviors by comparing simple execution rewards with more complex reward shaping.

Our goal is to understand whether explicit intermediate reasoning combined with execution-guided RL can improve the performance and robustness of small Text-to-SQL models.

2 Reading List

Foundational Reinforcement Learning

- Sutton, R. S., & Barto, A. G. *Reinforcement Learning: An Introduction*. MIT Press, 2018.
- Schulman, J. et al. "Proximal Policy Optimization Algorithms." *arXiv*, 2017.

Text-to-SQL and Semantic Parsing

- Yu, T. et al. "Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL." *EMNLP*, 2018.
- Zhong, V. et al. "Seq2SQL: Generating Structured Queries from Natural Language Using Reinforcement Learning." *arXiv*, 2017.

LLM-Based Text-to-SQL Systems

- Pourreza, M., & Rafiei, D. "DIN-SQL: Decomposed In-Context Learning of Text-to-SQL with Self-Correction." *NeurIPS*, 2023.
- Li, G. et al. "SQL-R1: Reinforcement Learning for Reasoning in Text-to-SQL." 2025.

Execution-Guided and RL-Based Training

- Arctic Team. "Arctic Text-to-SQL: Execution-Guided Reinforcement Learning for Structured Query Generation." 2025.
- Yu, W. et al. "Execution-Guided Decoding for Semantic Parsing." *ACL*, 2018.

3 Key Directions and Expected Outcomes

Direction 1: Intermediate Sketch Representation

We will design a structured sketch representation extracted from SQL queries (tables, columns, joins, aggregation, filters). The goal is to determine whether generating this intermediate plan improves the model's reasoning ability compared with direct SQL generation.

Direction 2: Execution-Guided Reinforcement Learning

We will apply GRPO-based reinforcement learning with a simple execution reward derived from database query results. This allows the model to improve through interaction with the environment without complex reward engineering.

Direction 3: Comparison of Training Strategies

We will systematically compare:

- (1) SFT vs RL-based post-training
- (2) Direct SQL generation vs sketch-based generation

This will allow us to analyze whether RL benefits more from structured intermediate reasoning.

Direction 4: Reward Design and Robustness

We will investigate whether simple execution rewards are sufficient, and whether more complex reward shaping leads to reward hacking or brittle behavior.

4 Expected Outcomes

We expect that:

1. Sketch-based decomposition will improve the reasoning ability of small Text-to-SQL models.
2. GRPO-based RL will further improve execution accuracy compared to SFT alone.
3. RL will be more effective when the model has an explicit intermediate reasoning structure.
4. Simple execution-based rewards will be more robust than heavily engineered reward functions.

These results will help clarify the role of **intermediate reasoning structures and reinforcement learning in improving Text-to-SQL systems**, particularly for small language models.